

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	27 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

Step 1: Problem Statement

Selected Problem

Farmers face challenges in selecting the most suitable crop because they find it difficult to analyze soil nutrients and environmental conditions. As a result, crop selection is often based on experience rather than data, leading to lower productivity and inefficient resource utilization.

Step 2: Brainstorming

Idea Category 1 – Crop Recommendation

- Smart crop recommendation based on soil nutrients
 - Crop suitability prediction
 - Soil nutrient analysis (N, P, K)
 - Environmental condition analysis
 - Crop productivity optimization
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Idea Category 2 – Data Analysis

- Data preprocessing and cleaning
 - Soil and environmental parameter analysis
 - Environmental data analysis
 - Feature selection
 - Dataset exploration and analysis
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Idea Category 3 – Machine Learning

- Machine learning classification model
 - Model training and evaluation
 - Crop prediction model
 - Prediction accuracy analysis
 - Machine learning model optimization
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Idea Category 4 – Web Application

- Soil and environmental data entry
 - Crop recommendation interface
 - Input validation
 - Crop recommendation results
 - User-friendly web application
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Idea Category 5 – Agricultural Decision Support

- Data-driven crop selection
 - Soil health insights
 - Crop recommendation support
 - Improved agricultural productivity
 - Sustainable farming decision support
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Step 3: Idea Prioritization

Idea	Importance	Feasibility	Priority
Smart Crop Recommendation	Very High	High	☆☆☆☆☆
Soil Analysis	Very High	High	☆☆☆☆☆
Crop Suitability Prediction	High	High	☆☆☆☆
Environmental Analysis	High	High	☆☆☆☆
Machine Learning Classification Model	High	Medium	☆☆☆☆
Crop Prediction Model	Medium	Medium	☆☆☆

Agricultural Dashboard	Medium	High	☆☆☆
Crop Prediction Web Application	Medium	Medium	☆☆☆

Final Selected Ideas

- Smart Crop Recommendation System
- Soil and Environmental Analysis
- Machine Learning-Based Crop Prediction
- User-Friendly Web Application

These ideas form the core modules of the Smart Agricultural Production Optimization Engine. The system enables farmers to make data-driven crop selection decisions by analyzing soil nutrients and environmental parameters through a trained machine learning model deployed as a Flask-based web application.